RADIANT: A Radiomics-Aware Domain-Informed Adversarial Network for High-Grade to Low-Grade Brain Tumor Adaptation

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Abstract

Unsupervised domain adaptation (UDA) transfers knowledge from labeled source domains to unlabeled target domains, addressing labeling challenges in new domains. In medical imaging, UDA has been applied to brain tumor segmentation; however, most existing approaches do not address high-grade glioma (HGG) to low-grade glioma (LGG) adaptation, rely on naive distributional alignment, lack structural constraints, and limit effective adversarial alignment. To address this, we propose a Radiomics-Aware Domain-Informed Adversarial Network (RADIANT), a novel structure-preserving domain adaptation framework for HGG to LGG adaptation. It employs a lightweight 3D U-Net for hierarchical representation learning and a radiomics-guided consistency strategy that links deep representations to global tumor characteristics to preserve anatomical integrity. It also introduces a dual-mode consistency mechanism and uses adversarial domain alignment to minimize discrepancies across domains. The proposed model is evaluated on three publicly available Brain Tumor Segmentation (BraTS) datasets: BraTS 2018, 2019, and 2020. Across the evaluated adaptation settings, it achieves Dice Similarity Coefficient (DSC) scores ranging from 80.17% to 86.91% for Whole Tumor (WT), 54.73% to 57.75% for Tumor Core (TC), and 48.01% to 59.24% for Enhancing Tumor (ET) on BraTS 2018. On BraTS 2019, the corresponding DSC ranges are 82.77% to 89.77% (WT), 53.97% to 64.51% (TC), and 48.09% to 58.19% (ET). For BraTS 2020, the model achieves DSC scores ranging from 76.21% to 85.52% (WT), 52.91% to 58.36% (TC), and 47.08% to 53.80% (ET). Our method achieves balanced and robust subregion performance across BraTS datasets while maintaining computationally efficient HGG-to-LGG adaptation without requiring LGG annotations.

Keywords: Domain Adaptation; Adversarial Domain Alignment; Structure Preserving; High Grade Glioma; Low Grade Glioma

1 Introduction

Among the various tasks in medical image segmentation, brain tumor segmentation has attracted immense attention in the field, as medical image analysis continues to play an increasingly important role in modern healthcare [1, 2, 3]. Precise and consistent segmentation of brain tumors is crucial for accurate diagnosis, effective treatment planning, and reliable post-treatment evaluation [4, 5]. Gliomas represent a prevalent category of brain tumors, classified into specific grades according to histopathological and molecular attributes, with low-grade glioma (LGG) and high-grade glioma (HGG) getting the most thorough scientific attention [4]. Magnetic resonance imaging (MRI) is one of the most widely employed imaging methods both prior to and following surgery, and its multi-modality features are critical [6]. Common modalities include T1-weighted (T1), T1-weighted with contrast-enhanced (T1ce), T2-weighted (T2), and fluid-attenuated inversion recovery (FLAIR) [1].

Accurate automated segmentation not only facilitates precise lesion localization and morphological assessment but also offers objective and reproducible information to support preoperative planning and treatment monitoring [7]. However, in practice, segmentation findings do not always match expectations since images taken at different institutions may differ in terms of image-acquisition features as well as tumor placement, grade, and intensity [8]. Such variations can reduce the ability of segmentation models trained on both HGG and LGG images to generalize well [8]. Moreover, when training with multimodal MRI data, differences between imaging modalities, such as T2- and T1-weighted sequences, can lead to cross-modality domain shifts, causing significant performance degradation [8]. In these circumstances, Unsupervised domain adaptation (UDA) is valuable for brain tumor segmentation, as it allows convolutional neural networks (CNNs) to learn from labeled source-domain images while adapting to unlabeled target-domain data [8, 9].

Various domain adaptation methods [10, 11, 12, 4, 13, 1, 14, 8, 15, 16] have been proposed to improve brain tumor segmentation, while recent studies [17, 18, 19, 20] have also explored the use of radiomic features to enhance tumor analysis and segmentation performance. To address domain shifts, numerous domain-adversarial neural networks (DANN)-based frameworks have been developed, including teacher–student self-training frameworks, frequency-guided adaptation methods such as Cross-Modality and Cross-Disease Multi-source Unsupervised Domain Adaptation (C2-MUDA) [12], Frequency and Edge-aware U-Net (FE-Unet) [1], and source-free black-box adaptation techniques such as BBUDA [13]. These models learn domain-invariant representations and enable segmentation networks to generalize across source and target datasets without relying on target-domain annotations. In particular, several UDA-based frameworks include UDA-GS [11] for cross-center glioma segmentation, a self-ensembling adversarial framework [15] for semantic segmentation, and Fourier Visual Prompting (FVP) [16] for source-free adaptation. Another category of methods focuses on extracting radiomic features, such as variance-filtered prognostic frameworks [19] and 3D replicator neural networks [17], which are designed to capture key tumor characteristics from segmented regions.

Recent developments in domain adaptation have made progress; however, several challenges remain unaddressed. Many existing models [10, 11, 13] rely on naive distributional alignment, which can compromise clinically meaningful tumor geometry and limit generalization between HGG and LGG due to significant differences in tumor grade and intensity. Furthermore, existing approaches often fail to incorporate structural constraints that connect deep feature representations with global tumor characteristics, which are essential for preserving anatomical consistency. Their training strategies often lack mechanisms to control the influence of noisy predictions. Additionally, many approaches do not utilize adversarial domain alignment to bridge the gap between feature representations across different domains. Together, these limitations highlight the need for models that maintain anatomical and feature-level consistency while reducing domain-related variations in feature distributions.

This inspired the development of our proposed model, Radiomics-Aware Domain-Informed Adversarial Network (RADIANT), a novel structure-preserving domain adaptation framework that enables accurate brain tumor segmentation without requiring target-domain annotations. The architecture is based on a lightweight 3D U-Net backbone and incorporates a radiomics-guided consistency strategy together with adversarial domain alignment to improve tumor segmentation. Specifically, the framework adopts a dual-mode consistency mechanism in which the supervised mode learns from HGG annotations, while the unsupervised mode employs a self-feedback loop to extract radiomic features from the model's own predictions on LGG data. To further enhance robustness, an adversarial domain discriminator is integrated at the bottleneck, encouraging the encoder to learn domain-invariant representations. In addition, a lightweight radiomics regressor is incorporated to preserve anatomical consistency, enabling stable performance across different tumor grades. The proposed model is evaluated on three benchmark datasets: BraTS 2018, BraTS 2019, and BraTS 2020. Across the evaluated source-target adaptation settings, our model achieves DSC ranges of 80.17% to 86.91% (WT), 54.73% to 57.75% (TC), and 48.01% to 59.24% (ET) on BraTS 2018; 82.77% to 89.77% (WT), 53.97% to 64.51% (TC), and 48.09% to 58.19% (ET) on BraTS 2019; and 76.21% to 85.52% (WT), 52.91% to 58.36% (TC), and 47.08% to 53.80% (ET) on BraTS 2020. The contributions of our study are presented below:

- Proposes a novel structure-preserving unsupervised domain adaptation framework for 3D brain tumor segmentation that allows a model trained on HGG data to efficiently generalize radiomics consistency to LGG without the need for LGG annotations.
- Designs a lightweight 3D U-Net-based hierarchical feature learning architecture with instance normalization to improve domain robustness, extended skip connections for precise tumor boundary preservation, and a bottleneck that serves as a shared feature space for multiple tasks.
- Introduces a novel radiomics-guided consistency strategy that incorporates radiomics feature extraction along with a lightweight regression module connected to the bottleneck features. The module links deep feature representations to global tumor characteristics to maintain anatomical consistency.
- Proposes a novel dual-mode consistency mechanism that prevents early training from being affected by noisy predictions and progressively enforces stronger consistency.
- Incorporates adversarial domain **cor**relation **al**ignment (CORAL) to ensure domain invariance by reducing the discrepancy between feature distributions across different domains at the bottleneck. This helps prevent disruption of low-level

features that encapsulate modality-specific intensity patterns.

- Extensive evaluations have been conducted on three benchmark datasets (BraTS 2018, 2019, and 2020). The model has shown consistent subtumor segmentation performance across four MRI modalities.

The remainder of this paper is structured as follows. Section 2 reviews prior research on domain adaptation and radiomics feature extraction. Section 3 describes the methodology of the proposed model, including the problem formulation and overall architectural design. Section 4 presents the experimental results and findings of the proposed model. Section 5 discusses the significance of the findings and outlines the limitations. Finally, Section 6 concludes the paper by summarizing the main contributions and results.

2 Literature Review

In this section, we review prior research on domain adaptation methods and radiomics feature extraction.

2.1 Domain Adaptation

UDA techniques have been explored using various strategies such as GAN-based synthesis, frequency-domain-guided alignment, and adversarial training, as demonstrated in recent studies [10, 11, 12, 4, 13, 1, 14, 8, 15, 16]. To begin with, Hu et al. [10] proposed a UDA framework for brain structure segmentation from MRI volumes, employing a self-training teacher-student architecture with knowledge distillation to learn cross-domain representations. The framework integrated mutual-information-based alignment and output-map distillation with contrastive learning and was evaluated on the IBSR and MALC datasets. In the MALC to IBSR adaptation scenario, the method achieved a DSC of 84.01%, while in the IBSR to MALC adaptation, it achieved a DSC of 84.81%. Similarly, Hu et al. [11] proposed UDA-GS, a cross-center UDA framework for glioma segmentation, which uses GliomaMix for tumor–image composition and combined self-supervised regression, weighted consistency regularization, and mask fusion within a Mean-Teacher setup to adapt to new centers without local annotations. It achieved a DSC of 78.3% for WT, 72.3% for TC, and 69.0% for ET.

Furthermore, Chen et al. [12] introduced C2-MUDA, a frequency-guided UDA framework with prototype-aware contrastive learning for cross-modality and cross-disease brain lesion segmentation. It integrated Fourier-based augmentation with semantic contrastive learning to mitigate domain shift and enhance pseudo-labeling. The method achieved DSC of 78.48%, 79.61%, and 66.75% for the WT, TC, and ET, respectively. In another paper, Khalil et al. [4] developed a conditional GAN to synthesize multimodal MR images for HGG segmentation and later applied Fourier domain adaptation to align low-frequency styles with real data, thereby reducing domain shift. However, enabling segmentation algorithms trained on HGG data to generalize to LGG data is still largely unsolved due to significant domain differences between tumor grades.

In another study, Zheng et al. [13] proposed the Dual Domain Distribution Disruption with Semantics Preservation (DDSP) framework, which eliminated the need for target-style images and rendered the model domain-agnostic by employing semantic constraints to preserve anatomical structures. It achieved a DSC of 67.3%, 57.9%, and 80.7% for the T2 to T1, T2 to T1CE, and T2 to FLAIR segmentation tasks, respectively. Similarly, Lei et al. [1] developed FE-Unet, a lightweight brain tumor segmentation network that integrated frequency domain processing with edge preservation. They employed an edge-enhanced hybrid loss and a feature reconstruction wavelet encoder to improve segmentation performance. This method achieved average DSCs of 70.98%, 83.63%, and 74.60% on the BraTS 2018, BraTS 2019, and BraTS 2020 datasets, respectively. Similarly, Liu et al. [14] proposed BBUDA, a UDA framework that used a black-box source model without access to source data or model parameters. They introduced knowledge distillation to learn target-specific representations and applied unsupervised entropy minimization to regularize label confidence. It achieved DSCs of 82.14%, 31.02%, and 46.13% for the WT, ET, and TC, respectively, in the LGG-to-HGG adaptation task.

In another study, Qin et al. [8] proposed DS-Adv, a UDA framework for brain tumor segmentation, which combines dual-student and adversarial training techniques to address domain shifts in MR images. They further integrated a cross-coordination constraint on the target domain data to encourage the models to produce more confident predictions. It obtained DSCs of 88.85%, 48.31%, and 63.63% for the WT, TC, and ET, respectively, in the HGG-to-LGG adaptation task. Furthermore, Shanis et al. [15] proposed a UDA framework for semantic segmentation, UDA-SE, which combines self-ensembling with adversarial training to address domain shifts in MR images. This framework consisted of three modules: a student segmentation network, a teacher segmentation network, and a discriminator. It achieved DSCs of 85.80%, 66.43%, 67.11%, and 78.23% for the WT, TC, ET, and overall segmentation in HGG, and 84.11%, 32.67%, 47.11%, and 62.17% for the corresponding metrics in LGG. Finally, Wang et al. [16] introduced FVP for source-free UDA in medical image segmentation. FVP adapted a frozen source model to a target domain by attaching a learnable frequency-domain visual prompt to the target inputs and optimizing it against pseudo-labels via segmentation loss. It reported DSC values of 69.7% for the FLAIR to T2 domain shift and 81.3% for the T2 to FLAIR domain shift.

2.2 Radiomics Feature Extraction

Recent studies [17, 18, 19, 20] have explored the use of radiomic features in deep learning and machine learning (ML) frameworks for tumor classification, survival prediction, and outcome prediction. Initially, Rastogi et al. [17] developed a deep learning model that uses radiomics features from segmented glioma areas to predict patient survival, finding critical imaging aspects using a 3D replicator neural network. However, their approach does not directly incorporate a structural constraint that aligns deep visual representations with clinically meaningful tumor characteristics.

Furthermore, Sun et al. [18] developed a radiomics-based framework for glioma subclass classification and explored the relationship between tumor morphology and pathological features in adult-type diffuse glioma. This method resulted in an average DSC of 88.4% for the TC and 88.9% for WT. Moreover, Gong et al. [19] introduced a radiomics-based prognostic framework that used variance filtering and LASSO-guided feature selection, followed by XGBoost modeling, to provide imaging-generated prediction scores that were evaluated by Kaplan-Meier analysis. Nevertheless, radiomic features in their framework are used solely for prognostic prediction and are not integrated into a lightweight regression module that bridges deep feature representations with global tumor descriptors. Similarly, Urbanos et al. [20] presented ML algorithms using the radiomic features of brain tissue and hemorrhage on CT scans, alone or in combination with clinical data, to predict the outcome of subarachnoid hemorrhage (SAH). However, their method lacks a self-feedback mechanism that iteratively improves the feature representations by producing supervisory signals from its own predictions.

Our proposed approach RADIANT overcomes key limitations of existing studies [10, 11, 12, 4, 13, 1, 14, 8, 15, 16, 17, 18, 19, 20] with a novel, lightweight, and efficient model. We present a novel structure-preserving domain adaptation framework for accurate brain tumor segmentation without target-domain annotations. It combines a segmentation network, a radiomics regression module, and a domain discriminator. Built on a lightweight 3D U-Net backbone, the architecture integrates a radiomics-guided strategy linking bottleneck features to tumor characteristics to preserve anatomical consistency and employs adversarial domain alignment to ensure domain invariance. A novel dual-mode consistency mechanism is also included to prevent early training from being influenced by noisy predictions and gradually strengthen consistency over time.

3 Methodology

The objective of this study is to develop a domain adaptation framework that enables a 3D brain tumor segmentation model trained on high-grade tumor data to generalize effectively to low-grade data through radiomics-guided self-supervised learning and adversarial domain alignment. The proposed framework (Figure 1) consists of three interconnected components: a segmentation network, a radiomics regression module, and a domain discriminator. Additionally, it ensures that the model uses complementary supervision signals: ground-truth annotations from HGG guide the segmentation boundaries, radiomics features enforce anatomical consistency, and adversarial training aligns the feature distributions across domains.

3.1 Dataset

We evaluate our framework using three BraTS benchmark datasets from 2018 to 2020 [21, 22, 23], partitioned into two clinically distinct grades: HGG and LGG.

Table 1: Overview of the publicly available BraTS datasets (2018–2020) and tumor characteristics based on tumor grades. Abbreviations: Min/Abs = Minimal/Absent, ✓= Present, ✗= Absent.

Category	Attribute	LGG	HGG
Samples	BraTS2018	75	210
	BraTS2019	76	259
	BraTS2020	76	293
	Total	227	762
Properties	Contrast Enh.	Min/Abs	Intense
	Boundary	Diffuse	Well-defined
	Necrosis	✗	✓
	Edema	Min/Abs	Substantial
	WHO Grade	I–II	III–IV

The two grades. HGG tumors correspond to WHO grades III-IV and show radiologically distinct features, including intense contrast enhancement due to blood-brain barrier disruption, well-defined tumor boundaries, heterogeneous internal architecture with necrosis, and substantial peritumoral edema. On the other hand, LGG tumors correspond to WHO grades I-II and present fundamentally different characteristics, such as minimal to absent contrast enhancement, diffuse and poorly defined margins, homogeneous texture, and insignificant signal abnormalities that blend with normal brain parenchyma.

For both grades, all cases include four co-registered MRI sequences: FLAIR, T1-weighted (T1), T1-weighted with contrast enhancement (T1ce), and T2-weighted (T2) imaging. Ground-truth segmentation masks annotate three hierarchical tumor regions where WT encompasses all tumor tissues, TC excludes edema, and ET identifies regions of active tumor growth. Table 1 overviews the dataset information.

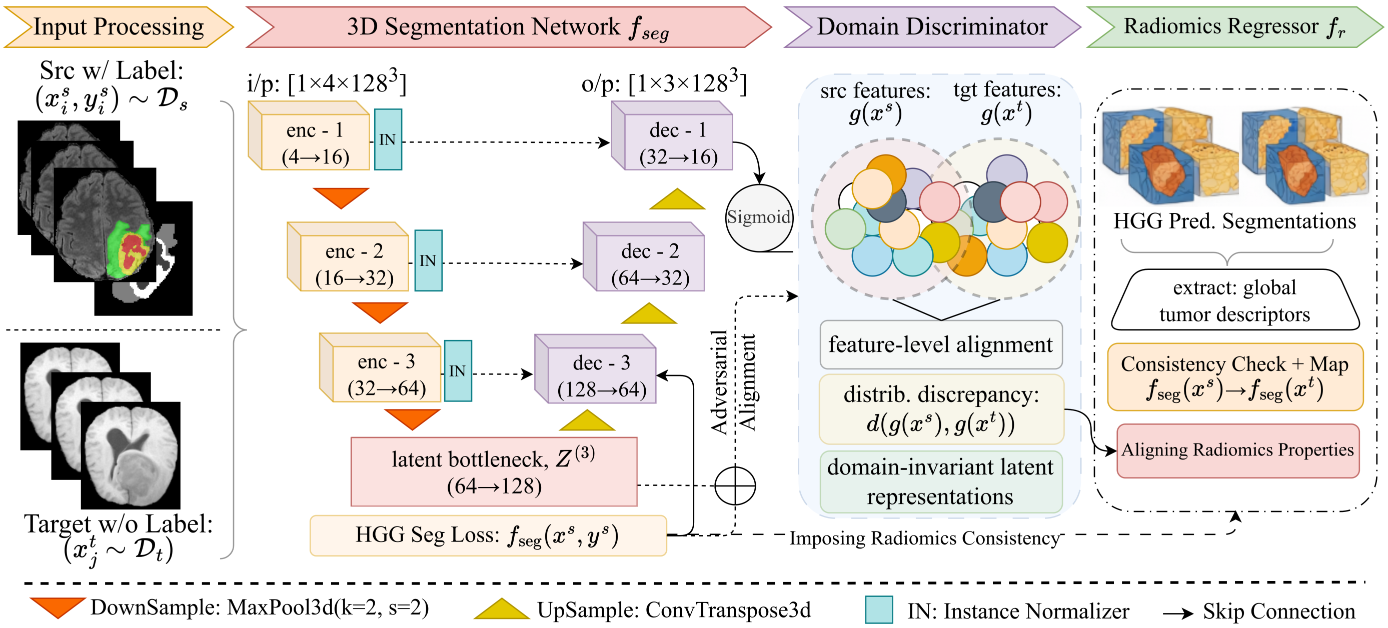

Figure 1: Overall framework of the radiomics-guided UDA method. Labeled source-domain volumes (x^s, y^s) and unlabeled target-domain volumes x^t are processed by a shared 3D U-Net f_{seg}. The encoder produces bottleneck features $Z^{(3)}$, which serve as the shared latent space. The supervised segmentation loss $\mathcal{L}_{\text{seg}}$ is applied to source samples, while a domain discriminator D enforces adversarial alignment. The radiomics regressor f_r introduces a global structure-preserving consistency loss.

3.2 Problem formulation

Most existing domain adaptation approaches address the domain discrepancy problem by aligning source and target feature distributions. However, in medical imaging, due to tumor-grade shift, naïve distributional alignment can compromise clinically meaningful tumor geometry. To address this limitation, we formulate domain adaptation as a structure-preserving learning problem, in which domain invariance is enforced while explicitly constraining tumor-level properties of the predictions.

We let $\mathcal{D}_s$ denote the source domain with data pairs (x_i^s, y_i^s) and $\mathcal{D}_t$ is the target domain with x_j^t, where $p_s(x, y) \neq p_t(x, y)$ due to systematic differences in tumor appearance and structure. The goal is to learn a segmentation function:

$$f_{\text{seg}} : \mathbb{R}^{C\times D\times H\times W} \rightarrow [0,1]^{3\times D\times H\times W}$$

that performs accurately on the target domain without access to target-domain annotations. Specifically, we consider the following constrained optimization objective below:

$$\begin{aligned} \min_{f_{\text{seg}}} \quad & \mathbb{E}_{(x^s,y^s)\sim\mathcal{D}_s}\Big[\mathcal{L}_{\text{seg}}(f_{\text{seg}}(x^s), y^s)\Big] \\ \text{s.t.} \quad & d\Big(g(x^s), g(x^t)\Big) \leq \epsilon \\ & \phi\Big(f_{\text{seg}}(x^t)\Big) \sim \phi(y^s) \end{aligned} \tag{1}$$

where $\epsilon > 0$ denotes a strict tolerance threshold bounding the latent domain discrepancy, $\sim$ represents a statistical similarity alignment operator, $g(\cdot)$ denotes the encoder of the segmentation network, $d(\cdot,\cdot)$ measures the discrepancy between source and target latent representations, and $\phi(\cdot)$ maps a segmentation and its corresponding image volume to a vector of global tumor characteristics and captures structural and intensity-related properties. The parameter ϵ explicitly controls the tight enforcement of feature invariance, while $\sim$ dictates the distribution matching criteria between target predictions and source ground truths. The first constraint promotes domain-invariant representations, while the second constrains adaptation using weak global tumor descriptors learned from the source domain and prevents degenerate solutions during feature alignment.

Unlike conventional UDA methods that rely solely on feature alignment, this formulation explicitly constrains adaptation by structural fidelity and enables reliable cross-grade segmentation without target-domain annotations. Notably, the structural constraint does not enforce voxel-level correspondence but regularizes the adaptation process using low-dimensional global priors that capture coarse tumor morphology.

3.3 Hierarchical Feature Learning

3.3.1 3D U-Net Architecture

We implement the segmentation backbone as a symmetric, lightweight, low-parameter 3D U-Net [24] with three downsampling-upsampling levels specifically designed for volumetric brain MRI processing. The network begins with four input channels corresponding to the four multimodal MRI sequences. Each encoding block consists of two consecutive 3×3×3 convolutions followed by instance normalization and LeakyReLU activation, with max pooling (2×2×2 stride) between levels to progressively increase receptive field while reducing spatial dimensions. Table 2 details the segmentation module's channel information.

The encoder expands feature dimensions through three hierarchical stages: starting with 16 channels at the first level to capture local texture patterns, expanding to 32 channels at the second level to integrate regional structural information, and reaching 64 channels at the third level to encode global tumor morphology. The bottleneck operates at 128 channels, with spatial dimensions reduced to one-eighth of the original input, creating a compressed yet information-rich representation space.

The decoder mirrors this structure through transposed convolutions with a stride of 2 for learnable upsampling, gradually recovering spatial resolution while reducing channel dimensions. The final layer employs a 1×1×1 convolution to produce three output channels corresponding to WT, TC, and ET probability maps, with sigmoid activation providing per-voxel probability estimates. The complete segmentation network contains only 1.4M parameters (approx.) optimized for 3D brain tumor segmentation.

3.3.2 Instance Normalization Design

We utilize instance normalization throughout the architecture to enhance domain robustness. Instance normalization computes mean and variance statistics independently for each sample rather than across the batch. For a feature map $X \in \mathbb{R}^{C\times D\times H\times W}$, the operation is defined as shown in Eq. (2):

$$\mathrm{IN}(X)_{c,i} = \gamma_c \left(\frac{X_{c,i} - \mu_c(X)}{\sqrt{\sigma_c^2(X) + \zeta}} \right) + \beta_c \tag{2}$$

In this formulation, $\mu_c(X)$ and $\sigma_c^2(X)$ represent the mean and variance computed per channel c across all spatial positions i within the sample, where the total number of spatial positions is $N = D \times H \times W$. The parameters $\gamma_c, \beta_c \in \mathbb{R}$ are learnable channel-wise scale and shift terms that allow the network to recover the original activation distribution when beneficial, while $\zeta > 0$ ensures numerical stability by preventing division by zero.

This design is critical for domain adaptation between HGG and LGG data, where intensity distributions differ substantially due to biological variations and acquisition protocols. Instance normalization prevents cross-domain statistical contamination during training by ensuring each sample is normalized independently. To understand its effect, we consider the gradient propagation through the following Equation (3):

$$\frac{\partial \mathrm{IN}(X)_{c,i}}{\partial X_{c,j}} = \frac{\gamma_c}{\sqrt{\sigma_c^2 + \zeta}} \Bigg(\delta_{ij} - \frac{1}{N} \\ - \frac{(X_{c,i} - \mu_c)(X_{c,j} - \mu_c)}{N(\sigma_c^2 + \zeta)} \Bigg) \tag{3}$$

Here, δ_{ij} denotes the Kronecker delta, where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. It acts as the indicator function during gradient computation. This gradient structure ensures stable backpropagation while enforcing domain invariance through sample-wise statistical whitening. Unlike batch normalization, which introduces dependencies between samples through batch statistics, instance normalization maintains consistent behavior regardless of batch size. Thus, we eliminate a major source of domain-specific bias in feature representations by decoupling normalization statistics from other samples.

3.3.3 Skip Connections for Boundary Preservation

To maintain precise tumor boundary localization despite domain shift and resolution reduction, we extended the skip connections [25] at each resolution level. These connections preserve fine-grained spatial information that would otherwise be lost during downsampling. At each decoding level l, the upsampled decoder feature map is fused with the encoder feature map from the same spatial resolution via channel-wise concatenation, followed by a learnable convolutional transformation:

$$Z^{(l)}_{\text{fused}} = \mathcal{F}^{(l)} \left(\left[Z^{(l)}_{\text{enc}}, U^{(l)} \right] \right) \tag{4}$$

where $[\cdot,\cdot]$ denotes channel-wise concatenation, $Z^{(l)}_{\text{enc}}$ and $U^{(l)}$ represent the encoder and upsampled decoder tensor feature maps at layer level l, and $\mathcal{F}^{(l)}$ represents a functional decoder mapping operator comprising a convolutional block. This block integrates high-resolution boundary cues from early encoding layers with semantically rich contextual information from deeper decoder layers.

This fusion mechanism enables the decoder to construct a multi-scale representation that is essential for accurate tumor segmentation across varying tumor grades and imaging conditions. From a representational perspective, skip connections preserve high-frequency spatial information that is essential for delineating diffuse tumor boundaries, particularly under domain shift. This multi-scale feature fusion allows the decoder to combine local boundary cues with global contextual information.

Table 2: Architecture of the segmentation block. Instance Normalization (IN) normalizes each sample individually across the channels to improve training stability and convergence.

Module Type	Input Channels	Output Channels	Kernel Size	Parameters
Encoder Block 1	Conv3d×2 + IN	$4 \to 16$	3×3×3	8,704
MaxPool 1	MaxPool3d	$16 \to 16$	2×2×2	0
Encoder Block 2	Conv3d×2 + IN	$16 \to 32$	3×3×3	41,536
MaxPool 2	MaxPool3d	$32 \to 32$	2×2×2	0
Encoder Block 3	Conv3d×2 + IN	$32 \to 64$	3×3×3	166,016
MaxPool 3	MaxPool3d	$64 \to 64$	2×2×2	0
Bottleneck	Conv3d×2 + IN	$64 \to 128$	3×3×3	663,808
Upsample 3	ConvTranspose3d	$128 \to 64$	2×2×2	5,600
Decoder Block 3	Conv3d×2 + IN	$128 \to 64$	3×3×3	131,904
Upsample 2	ConvTranspose3d	$64 \to 32$	2×2×2	6,416
Decoder Block 2	Conv3d×2 + IN	$64 \to 32$	3×3×3	83,008
Upsample 1	ConvTranspose3d	$32 \to 16$	2×2×2	4,112
Decoder Block 1	Conv3d×2 + IN	$32 \to 16$	3×3×3	20,768
Output Layer	Conv3d	$16 \to 3$	1×1×1	151

3.3.4 Bottleneck as Shared Representation

The bottleneck serves as a shared representation space where features converge before being distributed to task pathways, as detailed in Algorithm 1. Additionally, it forces the encoder to learn a unified representation that simultaneously satisfies segmentation accuracy, radiomics predictability, and domain invariance [26]. We formulate this multi-objective learning (see Eq. (5)) through a composite loss function applied to the bottleneck features $Z^{(3)}$.

$$\begin{aligned}\mathcal{J}(Z^{(3)}) = \mathbb{E}\Big[\ell_{\text{seg}}\Big(f_s(Z^{(3)}), Y\Big)\Big] \\ + \lambda_r\,\mathbb{E}\Big[\Big\|f_r(Z^{(3)}) - R\Big\|^2\Big] \\ + \lambda_a\,\mathbb{E}\Big[d_{\mathcal{H}\Delta\mathcal{H}}\Big(Z_s^{(3)}, Z_t^{(3)}\Big)\Big]\end{aligned} \tag{5}$$

In this formulation, ℓ_{seg} represents the segmentation loss comparing predictions $f_s(Z^{(3)})$ against ground truth masks Y. The radiomics consistency term measures the squared error between predicted radiomics features $\phi(f_r(Z^{(3)}))$ and actual radiomics signatures R. The domain alignment term $d_{\mathcal{H}\Delta\mathcal{H}}$ quantifies the distribution divergence between source and target bottleneck features. The coefficients λ_r and λ_a balance these competing objectives.

The composite objective in Eq. (5) is applied only at the bottleneck layer, which serves as the shared representation space across all tasks. Constraining adaptation at this level avoids forcing strict alignment of early-layer features, which are known to encode domain-specific intensity statistics in medical images. The bottleneck achieves substantial dimensionality reduction through spatial compression while expanding semantic capacity through channel expansion. This compression forces the network to discard spatially redundant information while preserving semantically meaningful features. The bottleneck thus acts as an information filter. It retains only those features that contribute to segmentation accuracy, radiomics predictability, and domain invariance.

3.4 Radiomics Consistency Bridge

We introduce radiomics consistency as a novel structural constraint that bridges the semantic gap between deep visual features and tumor characteristics. Figure 2 visualizes the radiomics consistency alignment process.

3.4.1 Radiomics Feature Extraction

Radiomics features are quantitative imaging biomarkers extracted from segmented tumor regions that capture tumor morphology, intensity statistics, and texture patterns [27]. We employ PyRadiomics v3.0.1[1] to compute these features through first-order statistics, shape descriptors, and texture features from tumor masks [28]. The extractor operates at native voxel spacing without resampling and uses a fixed bin width of 25. Features are restricted to the original image space. This excludes filter banks such as wavelet or Laplacian-of-Gaussian transforms. The extraction process transforms a binary segmentation mask M and corresponding image intensity volume I into a feature vector $r \in \mathbb{R}^{72}$ through a mapping function ϕ. We express the feature vector as shown in Equation (6):

$$r = \phi(I, M) = [\phi_{\text{shape}}(M), \phi_{\text{first}}(I, M), \phi_{\text{texture}}(I, M)] \tag{6}$$

where ϕ_{shape} computes 14 geometric shape descriptors from M, including elongation, sphericity, surface area, and axis

[1] https://github.com/AIM-Harvard/pyradiomics/releases/tag/v3.1.0

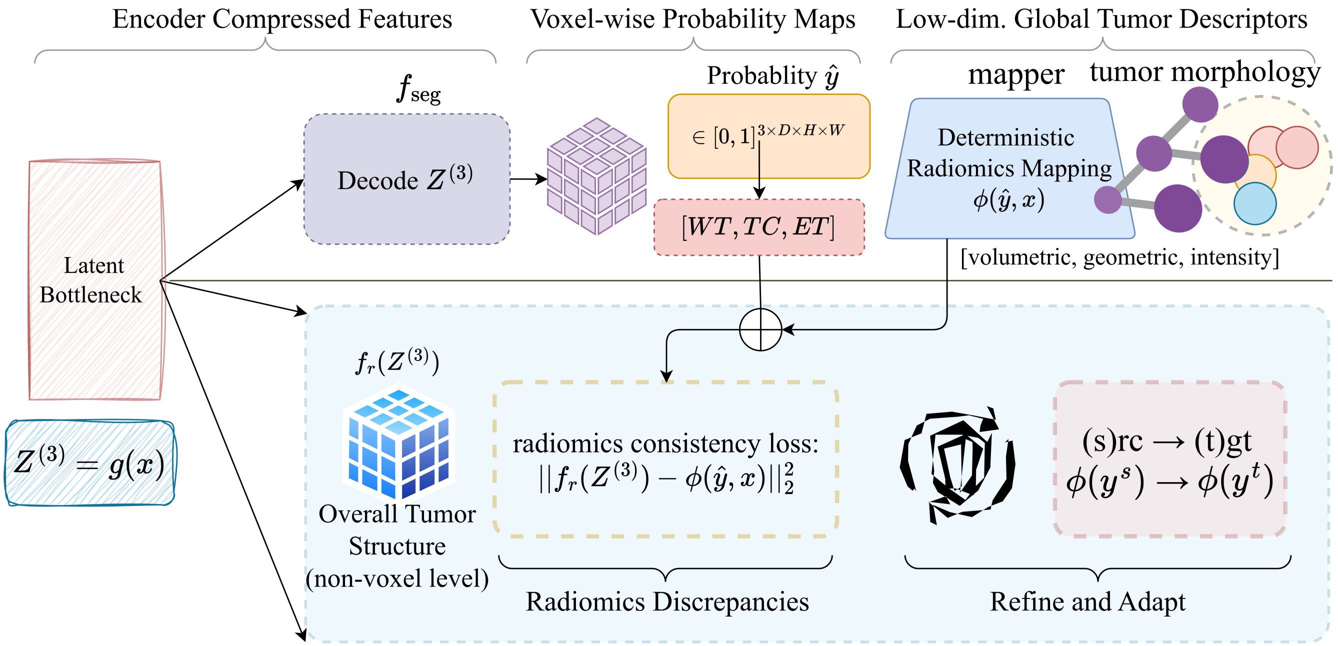

Figure 2: Radiomics-consistent UDA. The network encodes input images into a shared latent representation, which is decoded into voxel-wise tumor predictions and simultaneously passed through a radiomics regressor to predict global tumor descriptors. A deterministic mapping extracts radiomics features from the segmentation, and a consistency loss aligns the two pathways, shaping the latent space to preserve plausible tumor structure and guide adaptation in the target domain without voxel-level supervision.

lengths; ϕ_{first} calculates 18 intensity statistics within the tumor region, including mean, variance, skewness, kurtosis, energy, and entropy; and ϕ_{texture} extracts 40 texture descriptors comprising Gray Level Co-occurrence (24) and Run Length (16) Matrix features. Together, these produce $d = 72$ features for each HGG case.

The feature key set is fixed from the first source-domain case and held constant across all subsequent cases; missing features for any case are set to zero. Thus, for all source-domain cases, features are precomputed once before training begins and cached to disk. The extraction step ϕ is therefore non-differentiable and operates outside the gradient computation graph. For target-domain cases, ground-truth annotations are unavailable due to our defined problem formulation. Features are therefore extracted from the model's own segmentation predictions at each adaptation step.

We further compute geometric fallback features using computational geometry algorithms. This fallback produces four shape-based proxies: voxel volume, surface area, compactness, and eccentricity. It is invoked only when PyRadiomics raises an exception on degenerate masks. In this study, all radiomics features are normalized using source-domain statistics and are computed within fixed-size volumetric patches; this provides a consistent reference scale across training samples.

3.4.2 Radiomics Regressor

To bridge deep feature representations and global tumor descriptors, we introduce a lightweight radiomics regression module attached to the bottleneck features [29]. The module consists of a multilayer perceptron that maps the bottleneck representation to the target radiomics vector. The regression objective minimizes the squared error between predicted and ground-truth radiomics features computed from source-domain annotations. Importantly, the radiomics regression task acts as an auxiliary constraint rather than a primary supervision signal. Its role is to regularize the bottleneck representation by enforcing consistency with global tumor morphology. As a result, the regressor discourages feature collapse during domain alignment.

The radiomics regressor f_r learns to predict radiomics features directly from bottleneck representations. This creates a differentiable pathway between deep features and clinical descriptors. The bottleneck produces a feature map $Z \in \mathbb{R}^{128\times D'\times H'\times W'}$. Global average pooling reduces this to a 128-dimensional vector, which is then projected through a two-layer multilayer perceptron, defined as shown in Equation (7):

$$f_r(Z) = W_2 \cdot \sigma(W_1 \cdot \text{GAP}(Z) + b_1) + b_2 \tag{7}$$

where, $W_1 \in \mathbb{R}^{64\times 128}$ and $W_2 \in \mathbb{R}^{72\times 64}$ are learnable weight

Algorithm 1 Three-Component Domain Adaptation with Bottleneck Multi-Objective Learning

Input: source (X_s, Y_s), target X_t, encoder g, segmentation head f_{seg}, radiomics regressor f_r, domain discriminator D

Initialize: g, f_{seg}, f_r, D, weights λ_r, λ_a
for each training epoch **do**
  for each batch (X_s, Y_s, X_t) **do**
    // Forward pass through all components
    encode: $Z_s \leftarrow g(X_s)$, $Z_t \leftarrow g(X_t)$
    segment: $\hat{Y}_s \leftarrow f_{\text{seg}}(Z_s)$, $\hat{Y}_t \leftarrow f_{\text{seg}}(Z_t)$
    predict radiomics: $\hat{R}_s \leftarrow f_r(Z_s)$, $\hat{R}_t \leftarrow f_r(Z_t)$
    // Multi-objective learning at bottleneck
    segmentation loss: $L_{\text{seg}} \leftarrow \ell_{\text{seg}}(\hat{Y}_s, Y_s)$
    radiomics loss: $L_r \leftarrow \|\hat{R}_s - \phi(X_s, Y_s)\|^2$
    domain align: $L_a \leftarrow d_{\mathcal{H}\Delta\mathcal{H}}(Z_s, Z_t)$
    // Update discriminator (adversarial)
    compute $L_D \leftarrow -\log D(Z_s) - \log(1 - D(Z_t))$
    update D via gradient descent on L_D
    // Update encoder with GRL (reverse gradients)
    apply GRL to Z_s, Z_t with scaling factor λ
    encoder loss: $L_{\text{total}} \leftarrow L_{\text{seg}} + \lambda_r L_r + \lambda_a L_a$
    update g, f_{seg}, f_r via gradient descent on L_{total}
    // CORAL loss for second-order alignment
    compute covariance matrices C_s, C_t
    compute $L_{\text{coral}} \leftarrow \frac{1}{4d^2}\|C_s - C_t\|_F^2$
    update g with L_{coral}
  end for
end for

Output: adapted model f_{seg} for target domain

matrices, b_1 and b_2 denote their corresponding learnable bias vectors, $\text{GAP}(\cdot)$ signifies the Global Average Pooling operator, and $\sigma(\cdot)$ represents the element-wise ReLU activation function.

The regressor is trained to minimize the mean squared error between predicted and extracted radiomics features and can be expressed as the following Equation (8):

$$\mathcal{L}_r = \frac{1}{d}\sum_{i=1}^{d}(\hat{r}_i - r_i)^2 \tag{8}$$

where $d = 72$ is the radiomics feature (see Section 3.4.1) dimensionality, $\hat{r}$ denotes the predicted radiomics feature vector produced by the regressor from the latent representation Z, and r denotes the radiomics feature vector extracted from the input image and its corresponding mask: ground-truth annotations for source-domain samples and model predictions for target-domain samples.

Both feature vectors are normalized to zero mean and unit variance using statistics computed from the source domain. Here, the gradients do not flow through the extraction step ϕ. They propagate only through f_r, which ensures stable training without a differentiable radiomics pipeline. This encourages the encoder to learn representations that correlate with measurable tumor properties.

3.4.3 Dual-Mode Consistency

The radiomics consistency operates in two distinct modes depending on data availability and creates complementary supervision signals that guide adaptation at different learning stages (Algorithm 2). Specifically, the module dynamically changes constraints depending on the domain origin of the active input batch to actively preserve tumor geometry during cross-grade adaptation.

Algorithm 2 Dual-Mode Radiomics Consistency with Self-Feedback Refinement

Input: source (X_s, Y_s), target X_t, encoder g, segmentation head f_{seg}, radiomics extractor ϕ, radiomics regressor f_r, warm-up schedule $\beta(e)$

init g, f_{seg}, f_r, $\hat{Y}_t^{\text{prev}} \leftarrow \emptyset$, $\beta_{\text{self}} \leftarrow 0$
for each training epoch e **do**
  for each batch (X_s, Y_s, X_t) **do**
    // Forward pass
    encode: $Z_s \leftarrow g(X_s)$, $Z_t \leftarrow g(X_t)$
    segment: $\hat{Y}_t \leftarrow f_{\text{seg}}(Z_t)$
    // Supervised mode (source with GT)
    extract GT radiomics: $R_s^{\text{true}} \leftarrow \phi(X_s, Y_s)$
    predict radiomics: $\hat{R}_s \leftarrow f_r(Z_s)$
    supervised loss: $L_{\text{sup}} \leftarrow \|\hat{R}_s - R_s^{\text{true}}\|^2$
    // Self-supervised mode (target with predictions)
    binarize and extract: $R_t^{\text{pred}} \leftarrow \phi(X_t, \text{argmax}(\hat{Y}_t))$
    predict radiomics: $\hat{R}_t \leftarrow f_r(Z_t)$
    self-consistency: $L_{\text{self}} \leftarrow \|\hat{R}_t - R_t^{\text{pred}}\|^2$
    // Self-feedback refinement
    if $\hat{Y}_t^{\text{prev}} \neq \emptyset$ **then**
      compute quality: $q \leftarrow \text{IoU}(\hat{Y}_t^{\text{prev}}, \hat{Y}_t)$
      update weight: $\beta_{\text{self}} \leftarrow \beta(e) \cdot \beta_{\text{self}}^{\max} \cdot q$
    end if
    // Total radiomics loss
    $L_{\text{radio}} \leftarrow L_{\text{sup}} + \beta_{\text{self}} \cdot L_{\text{self}}$
    update f_r via gradient descent on L_{radio}
    // Feedback to encoder
    update g using gradient from L_{radio}
    store: $\hat{Y}_t^{\text{prev}} \leftarrow \hat{Y}_t$
  end for
end for

Output: Radiomics-aware encoder g

For source domain samples with ground-truth segmentation Y_s, we compute radiomics features directly from annotations and enforce supervised consistency, as defined in Equation (9):

$$\mathcal{L}_r^{\text{sup}} = \mathbb{E}_{(X_s, Y_s)}[\|f_r(Z_s) - \phi(X_s, Y_s)\|^2] \tag{9}$$

Here, the MLP regressor f_r maps latent features to the 72-dimensional reference feature space to ground the hidden layers in clinically valid anatomy. This establishes a reliable mapping between bottleneck features and clinically validated tumor descriptors, teaching the network what medically relevant features look like. For the targeted domain samples without annotations, we implement unsupervised consistency by extracting radiomics from the model's own predictions $\hat{Y}_t$. The consistency term is defined in Equation (10):

$$\mathcal{L}_r^{\text{self}} = \mathbb{E}_{X_t}[\|f_r(Z_t) - \phi(X_t, \text{argmax}(\hat{Y}_t))\|^2] \tag{10}$$

where $\hat{Y}_t$ is the predicted soft class probability map for target volume X_t, $\mathbb{E}_{X_t}$ is the target domain expectation, $\|\cdot\|^2$ is the squared L_2 norm, Z_t is the target latent feature tensor, f_r is the MLP regressor, ϕ is the PyRadiomics operator, and $\text{argmax}(\cdot)$ is the voxel-wise maximum selection operator used to binarize continuous probabilities into a hard segmentation mask. This creates a feedback loop where the model's segmentation quality directly influences the radiomics consistency signal.

To safeguard the network from degenerate alignment solutions during early training iterations, a dynamic self-feedback weight adjustment is introduced. The system tracks structural stability across iterations by evaluating the Intersection over Union (IoU) between the active prediction $\hat{Y}_t$ and its historical counterpart from the previous epoch $\hat{Y}_t^{\text{prev}}$, producing a stabilization scalar q. The unsupervised consistency weight β_{self} is then modulated dynamically as:

$$\beta_{\text{self}} = \beta(e) \cdot \beta_{\text{self}}^{\max} \cdot q \tag{11}$$

where $\beta(e)$ is the epoch-dependent warm-up factor and $\beta_{\text{self}}^{\max}$ is a preset scalar ceiling. This formulation ensures that when prediction outputs oscillate aggressively, the self-supervised adaptation signal is penalized heavily to ensure structural fidelity.

3.4.4 Self-feedback mechanism

We design a self-feedback mechanism that creates a closed-loop refinement cycle without requiring external annotations. This mechanism operates by using the model's own predictions to generate supervisory signals, which then guide further refinement of the feature representations.

The feedback process begins with the current bottleneck features Z_t producing segmentation predictions $\hat{Y}_t$, as described in Section 3.4.2. From these predictions, we extract radiomics features $\phi(X_t, \hat{Y}_t)$ that describe the geometric and intensity properties of the segmented tumor region. These extracted features are then compared against the radiomics features predicted directly from the bottleneck, $f_r(Z_t)$, (i.e., creating a consistency loss). This self-consistency objective seeks to minimize the discrepancy between the predicted and extracted radiomics, as described in Equation (10). The formulation then establishes a feedback loop where better segmentation yields more accurate radiomics extraction, which in turn provides clearer supervision for improving the bottleneck features.

To ensure stable training, we implement a warm-up schedule that gradually increases the influence of this self-supervised loss. The weighting coefficient β_{self} evolves according to Equation (12):

$$\beta_{\text{self}}^{(e)} = \min\left(1, \frac{e - e_{\text{start}}}{e_{\text{end}} - e_{\text{start}}}\right) \cdot \beta_{\text{self}}^{\max} \tag{12}$$

Here, e denotes the current training epoch index, with e_{start} and e_{end} defining the initial and final boundaries of the warm-up period, and $\beta_{\text{self}}^{\max}$ represents a pre-defined scalar constant bounding the maximum permissible weight assigned to the self-supervised consistency loop. The mechanism prevents the model from being constrained by noisy predictions during early training while progressively enforcing stronger consistency as segmentation quality improves. The feedback mechanism creates a virtuous cycle where improved segmentation leads to better radiomics extraction, which provides more accurate supervision signals for refining the feature encoder.

3.5 Domain-Invariant Learning

To enforce domain invariance, we needed to align the feature distributions of different domains [30]. To achieve this, we applied an adversarial objective that minimizes the discrepancy between the domains' bottleneck representations. The model then avoids destructive interference with low-level features (which encode modality-specific intensity patterns) because the alignment is restricted to the compressed bottleneck space.

3.5.1 Gradient Reversal Mechanism

We implement a gradient reversal layer (GRL) to facilitate adversarial domain alignment within a single forward-backward training pass [31]. The GRL operates as an identity transformation during forward propagation and allows features to flow unchanged to the domain discriminator. However, during backpropagation, it multiplies incoming gradients by a negative scaling factor, which effectively reverses their direction before they reach the encoder. GRL implements the following transformation (Eq. (13)) during backpropagation:

$$\frac{\partial \mathcal{L}_{\text{adv}}}{\partial Z} = -\alpha_{\text{grl}} \cdot \frac{\partial \mathcal{L}_{\text{adv}}}{\partial \text{GRL}(Z)} \tag{13}$$

where Z is the bottleneck feature tensor, $\mathcal{L}_{\text{adv}}$ is the domain adversarial loss, $\text{GRL}(Z)$ represents the layer's forward identity mapping, and α_{grl} is the dynamic scaling factor controlling gradient reversal magnitude to regulate domain confusion. This creates a minimax game within standard gradient-based optimization: the discriminator learns to distinguish between source and target features, while the encoder receives gradients that encourage it to produce features that confuse the discriminator. The parameter α_{grl} follows an increasing schedule throughout training, starting from zero and gradually reaching its maximum value.

3.5.2 Domain Discriminator Architecture

We design a domain discriminator that processes flattened bottleneck features through three fully-connected layers with decreasing dimensionality, implementing a hierarchical decision process for domain classification [32]. The architecture is intentionally kept lightweight to prevent it from overpowering the primary segmentation task while remaining effective at distinguishing domain-specific patterns. The discriminator is trained using binary cross-entropy (BCE) loss to distinguish between source HGG and LGG feature distributions, following Equation (14):

$$\mathcal{L}_{\text{adv}}^{\text{disc}} = -\mathbb{E}_{Z_s}[\log D(Z_s)] - \mathbb{E}_{Z_t}[\log(1 - D(Z_t))] \tag{14}$$

Here, Z_s and Z_t represent bottleneck features from source and target domains, respectively, while $D(\cdot)$ outputs the probability of belonging to the source domain. Through gradient reversal, the encoder receives opposing gradients that encourage the production of domain-invariant features, creating the adversarial dynamic essential for adaptation.

3.5.3 Correlation Alignment

Beyond adversarial alignment of marginal feature distributions, we implement correlation alignment (CORAL) to match second-order statistics between domains [33]. This also aligns the feature covariance structures and ensures that not only individual feature means but also their interrelationships become domain-invariant. For source features Z_s and target features Z_t with batch size B and feature dimension d, we compute their covariance with Equation (15) and (16):

$$C_s = \frac{1}{B-1}(Z_s^\top Z_s - \frac{1}{B}(1^\top Z_s)^\top(1^\top Z_s)) \tag{15}$$

$$C_t = \frac{1}{B-1}(Z_t^\top Z_t - \frac{1}{B}(1^\top Z_t)^\top(1^\top Z_t)) \tag{16}$$

where 1 denotes a vector of ones. The coral loss (Eq. (17)) then minimizes the squared Frobenius norm between these covariance matrices:

$$\mathcal{L}_{\text{coral}} = \frac{1}{4d^2}\|C_s - C_t\|_F^2 \tag{17}$$

Here, the scaling factor $\frac{1}{4d^2}$ normalizes the loss by feature dimension.

4 Experiments and Results

4.1 Experimental setup

All cases are split 70% for training and 30% for evaluation for both of the domains. However, the domains serve fundamentally different roles across the two training phases. For instance, HGG cases form the labeled source domain. During Phase 1, the 70% HGG training partition provides ground truth masks for supervised pretraining. After adaptation in Phase 2, the 30% HGG evaluation partition is used to monitor whether the model retains source domain performance. LGG cases, by contrast, form the unlabeled target domain. During Phase 2, the 70% LGG training partition provides volumes for unsupervised adaptation, with ground truth masks withheld entirely throughout this process. The 30% LGG evaluation partition then serves as the primary benchmark, where ground truth masks are used for the first time solely to compute the DSC scores. Table 3 reports the data partitioning setup.

Table 3: Dataset partitioning across BraTS benchmarks. LGG training cases are used for unsupervised adaptation only, with no labels used during training.

Dataset	**Domain**	**Train**	**Eval**
BraTS 2018	HGG	147	63
	LGG	52	23
BraTS 2019	HGG	181	78
	LGG	53	23
BraTS 2020	HGG	205	88
	LGG	53	23

All input volumes are center-cropped and zero-padded to $128 \times 128 \times 128$ voxels, then normalized independently to zero mean and unit variance per modality. Following this, we stacked the four modalities (i.e., FLAIR, T1, T1ce, and T2) into a 4-channel input. Since BraTS volumes are already co-registered and skull-stripped, no resampling or skull stripping is applied. All experiments are implemented in PyTorch 2.6.0 and conducted on dual NVIDIA Tesla T4 GPUs (15 GB each) with 30 GB RAM under CUDA 12.4, running Python 3.11.13, with PyRadiomics v3.0.1 for feature extraction. The model is trained with a batch size of 2.

4.2 Evaluation on the datasets

A comprehensive summary of all quantitative results and detailed performance metrics is provided in Table 4.

On the BraTS 2018 dataset, the proposed model achieves a mean DSC of 83.12% for the WT subregion, with a consistency of 91.89%. In contrast, the TC subregion attains a lower mean DSC of 55.34% but demonstrates the highest

Table 4: Mean DSC values represent average segmentation accuracy, with Best and Worst indicating the range of performance variability. Range shows absolute variability, while Performance Spread quantifies this as a percentage of mean performance. Consistency indicates result stability (100 - Performance Spread), and (relative) Variability approximates the coefficient of variation based on range.

Dataset	Subregion	Best	Worst	Mean DSC	Range	Performance Spread (%)	Consistency (%)	Variability (%)
BraTS 2018	WT	0.8691	0.8017	0.8312	0.0674	8.11	91.89	2.03
	TC	0.5775	0.5473	0.5534	0.0302	5.46	94.54	1.36
	ET	0.5924	0.4801	0.5187	0.1123	21.65	78.35	5.41
	Avg.	0.6797	0.6097	0.6344	0.07	11.74	88.26	2.93
BraTS 2019	WT	0.8977	0.8277	0.8453	0.07	8.28	91.72	2.07
	TC	0.6451	0.5397	0.5762	0.1054	18.29	81.71	4.57
	ET	0.5819	0.4809	0.5494	0.101	18.38	81.62	4.6
	Avg.	0.7082	0.6161	0.6570	0.0921	14.98	85.02	3.75
BraTS 2020	WT	0.8552	0.7621	0.8198	0.0931	11.36	88.64	2.84
	TC	0.5836	0.5291	0.5421	0.0545	10.05	89.95	2.51
	ET	0.5380	0.4708	0.5089	0.0672	13.21	86.79	3.3
	Avg.	0.6589	0.5873	0.6236	0.0716	11.54	88.46	2.88

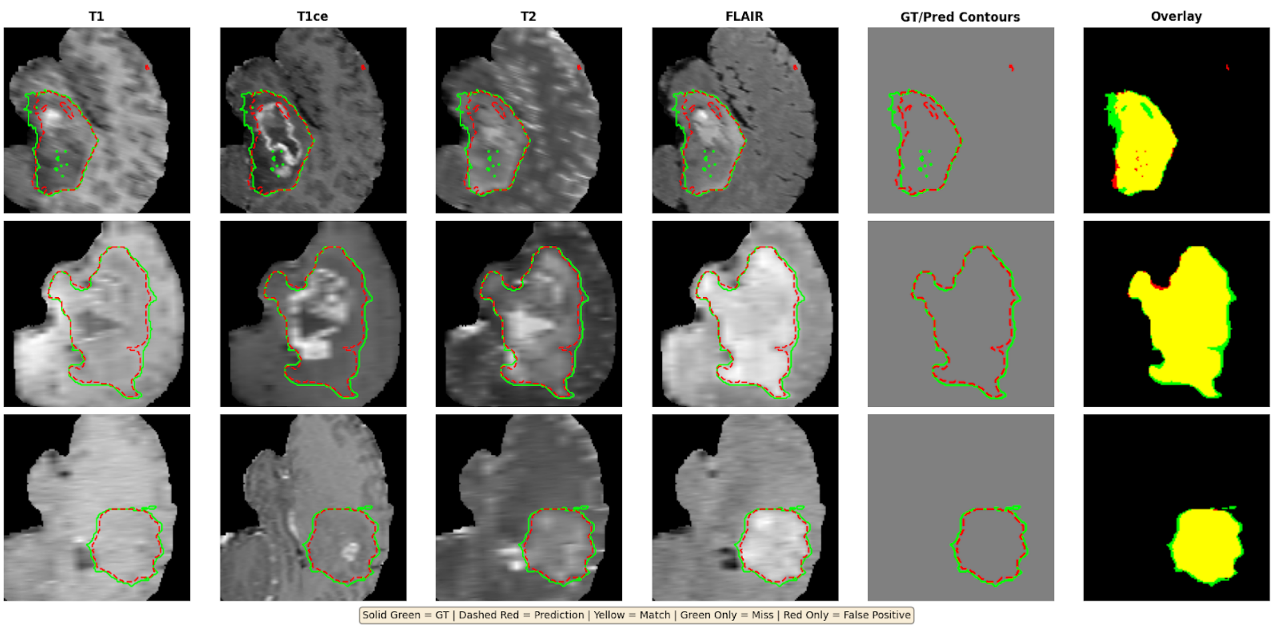

Figure 3: Visualization of the Whole Tumor (WT) region across four MRI modalities (T1, T1ce, T2, and FLAIR). The ground truth is shown in solid green and the predicted mask in dashed red. Overlapping regions indicate correct detections, green-only regions indicate missed tumor areas, and red-only regions indicate false positives.

stability among all reported results, with a consistency of 94.54%. The ET subregion remains the most challenging in this dataset, recording the lowest consistency of 78.35% and the largest performance spread of 21.65%. Overall, the BraTS 2018 dataset records an average mean DSC of 63.44% and a mean consistency of 88.26%.

For BraTS 2019, the model achieves its highest overall segmentation accuracy, with an average mean DSC of 65.70%. The WT subregion reaches the highest individual mean DSC across all experiments, at 84.53%. The TC and ET subregions show improved mean DSC of 57.62% and 54.94%, respectively, compared to 2018. However, both subregions show increased variability, with performance spreads of 18.29% and 18.38%, resulting in a slightly lower average consistency of 85.02% for this dataset.

On the BraTS 2020 dataset, the mean DSC for the WT, TC, and ET subregions are 81.98%, 54.21%, and 50.89%, respectively. Although the overall average mean DSC decreases to 62.36% compared to 2019, the results indicate improved stability. The average consistency is 88.46%, and the relative variability is the lowest among the three datasets at 2.88%. In addition, the ET subregion in 2020 shows a more controlled performance spread of 13.21% compared to the higher variability observed in the previous datasets.

4.3 Adaptation Performance on Subregions

Evaluation on the WT. The domain adaptation results for the WT subregion, illustrated in Figure 3 and summarized in Table 5, indicate that BraTS 2019 achieves the highest segmentation accuracy among all datasets.

For BraTS 2018 and BraTS 2020, the overall performance remains predominantly above 60%. For HGG, when the model is adapted using the T1 modality, it reaches its highest performance of 70.65% on the BraTS 2018 dataset, surpassing the other three modalities in the same configuration. A similar pattern is observed for the remaining modalities: T1ce achieves 71.48% when adapted with the T1ce modality, T2 reaches 69.82% when using the T2 modality, and Flair attains 72.12% when the Flair modality is targeted. These results demonstrate that the model achieves its highest segmentation accuracy when the source and target modalities align. This behavior is observed consistently across all datasets, highlighting the model's stable and robust domain adaptation performance for each MRI sequence.

Evaluation on the TC. Figure 4 visualizes the domain adaptation segmentation overlays for the TC subregion. Similar to the WT subregion, the advantage of within-modality adaptation is evident, where the model achieves optimal performance when the source and target modalities match. For this subregion, the T1ce modality provides the most effective adaptation, reaching its highest DSC of 49.54% on the BraTS 2019 dataset, followed by T1, Flair, and T2. The performance ranges for the different datasets are as follows: 38.75%–47.59% for BraTS 2018, 40.37%–49.54% for BraTS 2019, and 37.96%–46.69% for BraTS 2020. Table 6 reports the detailed results.

Evaluation on the ET. Table 7 presents the domain adaptation results for the ET subregion, as also presented in Figure 5. Among the three datasets, the highest overall segmentation performance is achieved on BraTS 2019, followed by BraTS 2018 and BraTS 2020. Consistent with the observations for the WT and TC subregions, the advantage of within-modality adaptation is evident, as the model achieves superior performance when the source and target modalities are identical. For the ET subregion, T1ce serves as the most effective adaptation modality, attaining its highest DSC of 44.60% on the BraTS 2018 dataset, followed by T1, T2, and Flair. In contrast, T2 records the lowest DSC values on BraTS 2019 and BraTS 2020, with values of 46.15% and 42.74%, respectively. The overall performance ranges from 36.31%–44.60% for BraTS 2018, 38.46%–47.25% for BraTS 2019, and 35.59%–43.78% for BraTS 2020.

4.4 Training Strategy and Hyperparameter Configuration

Training Strategy and Hyperparameter Configuration. The training is conducted in two phases: supervised pretraining on HGG data and unsupervised adaptation to the target domain. During the supervised phase, a StepLR scheduler with an initial learning rate of 1×10^{-3}, step size of 5, and $\gamma = 0.9$ is applied to ensure gradual convergence, while the radiomics weight is progressively increased from 0.1 to 0.2 to emphasize radiomics information after initial feature stabilization.

In the adaptation phase, distinct learning rate schedules are assigned to different network components to balance representation stability and domain-specific flexibility. The encoder is optimized with a lower learning rate of 1×10^{-4} using StepLR ($\gamma = 0.95$) to preserve pretrained representations, whereas the decoder employs a higher learning rate of 1×10^{-3} with the same decay factor to facilitate effective adaptation to the target domain. The discriminator operates with a fixed learning rate of 1×10^{-4} and $\beta = (0.5, 0.999)$ to maintain stable adversarial training.

Additionally, the gradient reversal factor α increases linearly from 1.0 to 2.0 over 200 epochs to progressively strengthen domain invariance, and radiomics consistency is introduced with a warmup strategy ($\lambda = 0$ for the first 10 epochs, then 0.0001) to prevent instability caused by early prediction noise. L2 regularization with weight decay $\omega = 1 \times 10^{-5}$ is applied throughout both phases to mitigate overfitting, as presented in Table 8.

Loss Components and Optimization Weights. The training framework integrates multiple loss functions to jointly optimize segmentation performance and domain adaptation, as summarized in Table 9. Initially, Dice loss is applied to both HGG and LGG segmentation tasks with per-region weighting and an optimization weight factor of 1.0. Following this, Focal loss, a BCE variant with hyperparameters $\alpha = 0.25$ and $\gamma = 2.0$, is employed on both domains with an optimization weight factor of 0.5 to mitigate class imbalance.

For the source domain (HGG), radiomics loss based on mean squared error is introduced, with its weight gradually increasing from 0.1 to 0.2 after 5 epochs to align deep feature representations with ground-truth radiomics. In the target domain (LGG), radiomics consistency, also using mean squared error, is enforced with a warmup strategy ($\lambda_{rad} = 0$ for the first 10 epochs, then 0.0001) to maintain self-consistency on pseudo-radiomics features. Subsequently, adversarial loss with $\lambda_{adv} = 0.001$ is applied to LGG, guiding the generator to produce features closely resembling those from HGG and thereby promoting domain-invariant representations. Finally, the discriminator loss, BCE applied across both HGG and LGG with weight $0.5\times(src+tgt)$,

Table 5: Domain adaptation results for the Whole Tumor (WT) subregion across the BraTS benchmark. The results are compared across individual MRI modalities (T1, T1ce, T2, and Flair) for both HGG and the domain-adapted LGG.

High Grade Glioma				BraTS2018				BraTS2019				BraTS2020			
T1	T1ce	T2	Flair	T1	T1ce	T2	Flair	T1	T1ce	T2	Flair	T1	T1ce	T2	Flair
✓	✗	✗	✗	0.7065	0.6816	0.6234	0.6401	0.7184	0.6935	0.6353	0.6520	0.6966	0.6717	0.6135	0.6302
✗	✓	✗	✗	0.6733	0.7148	0.6484	0.6651	0.6852	0.7267	0.6603	0.6770	0.6634	0.7049	0.6385	0.6552
✗	✗	✓	✗	0.5987	0.6033	0.6982	0.6899	0.6106	0.6243	0.7101	0.7018	0.5888	0.6232	0.6883	0.6801
✗	✗	✗	✓	0.6068	0.6317	0.6801	0.7212	0.6187	0.6436	0.6815	0.7382	0.5969	0.6218	0.6670	0.7016

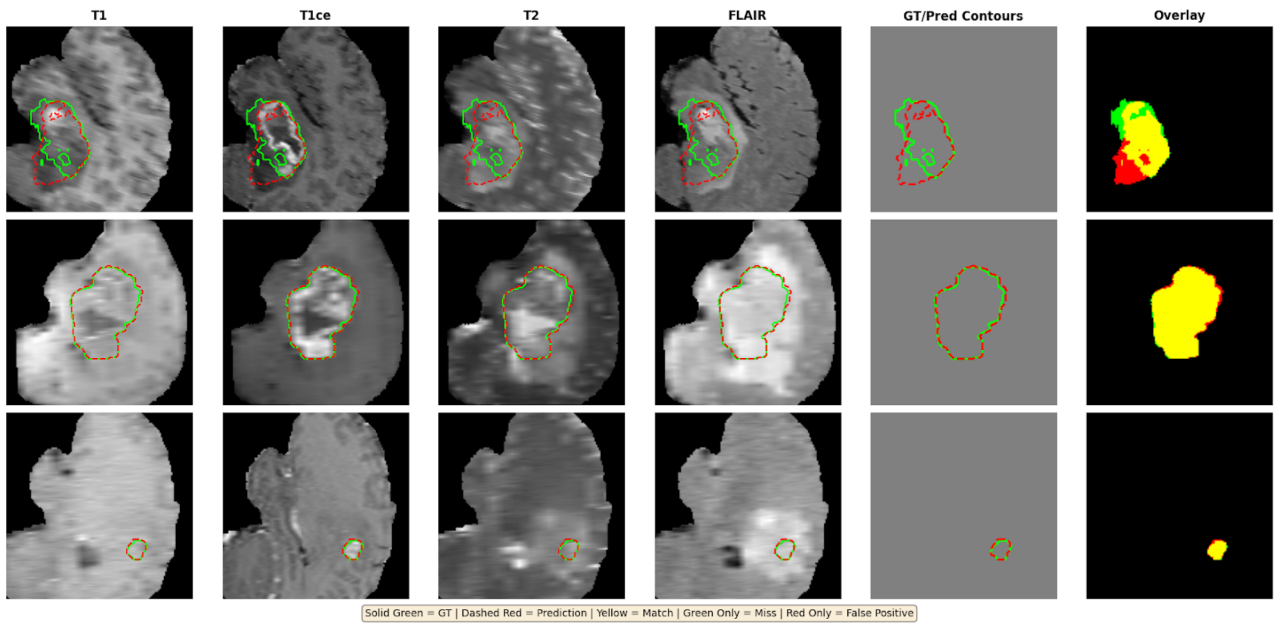

Figure 4: Qualitative visualization of segmentation across four MRI modalities. The figure highlights the Tumor Core (TC) regions, which encircle Enhanced Tumor (ET) and are surrounded by WT.

Table 6: DSC scores for the Tumor Core (TC) subregion across the BraTS benchmark per MRI modality. Adaptation between two identical domains/modalities provides better adaptation DSC results.

High Grade Glioma				BraTS2018				BraTS2019				BraTS2020			
T1	T1ce	T2	Flair	T1	T1ce	T2	Flair	T1	T1ce	T2	Flair	T1	T1ce	T2	Flair
✓	✗	✗	✗	0.4704	0.4538	0.4150	0.4261	0.4899	0.4733	0.4345	0.4456	0.4640	0.4423	0.4055	0.4166
✗	✓	✗	✗	0.4427	0.4759	0.4316	0.4237	0.4611	0.4954	0.4501	0.4414	0.4337	0.4669	0.4216	0.4327
✗	✗	✓	✗	0.3875	0.4095	0.4648	0.4531	0.4037	0.4268	0.4839	0.4725	0.3796	0.4016	0.4558	0.4426
✗	✗	✗	✓	0.3986	0.4206	0.4483	0.4716	0.4148	0.4379	0.4667	0.4896	0.3907	0.4127	0.4393	0.4609

Table 7: Enhancing Tumor (ET) DSC scores across the BraTS benchmark per MRI modality. The table compares cross-modal performance, highlighting the variability of the results when the model considers the specific modality marked with ✓. ET is the smallest and most challenging subregion, with minimal to absent contrast enhancement in LGG relative to HGG.

High Grade Glioma				BraTS2018				BraTS2019				BraTS2020			
T1	T1ce	T2	Flair	T1	T1ce	T2	Flair	T1	T1ce	T2	Flair	T1	T1ce	T2	Flair
✓	✗	✗	✗	0.4409	0.4254	0.3890	0.4004	0.4673	0.4515	0.4151	0.4265	0.4326	0.4271	0.3807	0.3921
✗	✓	✗	✗	0.4251	0.4460	0.4046	0.3951	0.4395	0.4725	0.4291	0.4195	0.4067	0.4378	0.3963	0.4163
✗	✗	✓	✗	0.3631	0.3838	0.4357	0.4263	0.3846	0.4072	0.4615	0.4505	0.3559	0.3766	0.4274	0.4171
✗	✗	✗	✓	0.3735	0.3942	0.4202	0.4319	0.3950	0.4177	0.4461	0.4670	0.3663	0.3810	0.4119	0.4532

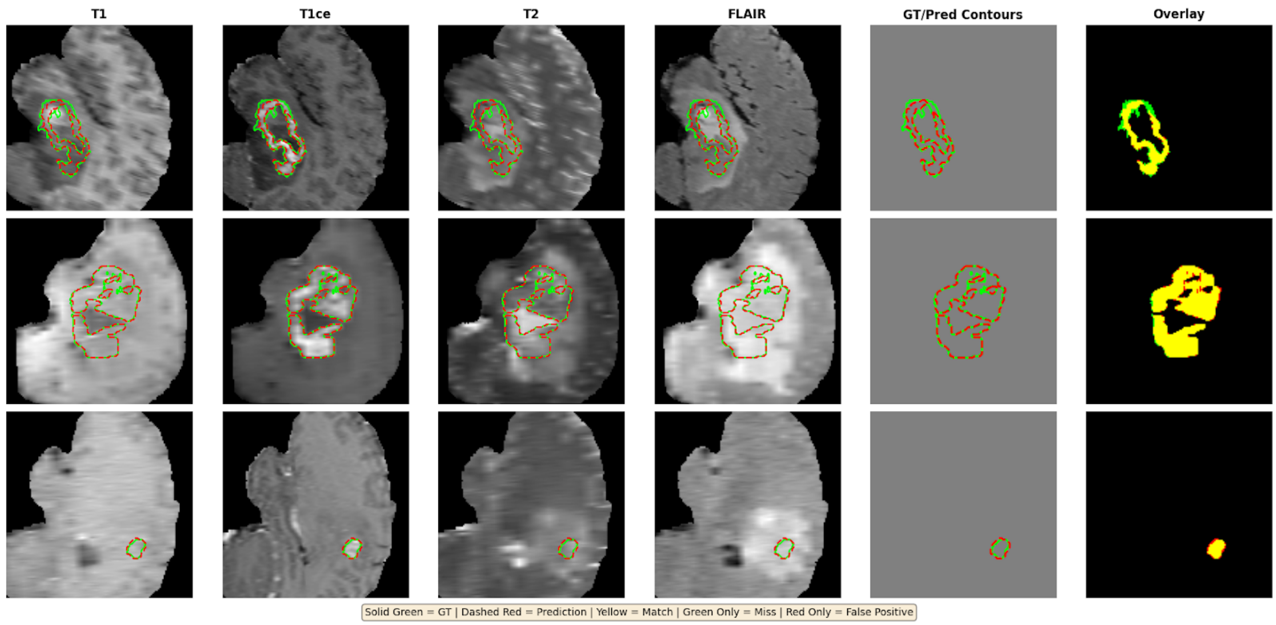

Figure 5: Visual comparison of Enhancing Tumor (ET) segmentation across four MRI modalities (T1, T1ce, T2, and FLAIR). ET is enclosed by both TC and WT, in this order. The figure highlights the agreement between GT annotations and model predictions.

Table 8: Hyperparameter and optimization settings for supervised pretraining and adaptation phases, including component-specific learning rates and regularization for domain-invariant learning.

Component	**Type**	**Parameters**	**Supervised**	**Adaptation**
LR Schedule	StepLR	step=5, γ=0.9, LR=1e-3	✓	✗
LR Schedule (Encoder)	StepLR	step=5, γ=0.95, LR=1e-4	✗	✓
LR Schedule (Decoder)	StepLR	step=5, γ=0.95, LR=1e-3	✗	✓
LR Schedule (Disc)	Fixed	LR=1e-4, β=(0.5, 0.999)	✗	✓
Radiomics Weight	Auto	0.1 (ep $\leq$ 5) $\rightarrow$ 0.2	✓	✗
Gradient Reversal α	Linear	1.0 $\rightarrow$ 2.0 over 200 eps	✗	✓
Weight Decay	L2 Reg	ω = 1e-5	✓	✓
Radiomics Consistency	Warmup	$\lambda = 0$ (ep $\leq$ 10) $\rightarrow$ 1e-4	✗	✓

Table 9: Loss components, their types, applied domains, and corresponding weights employed in the two-phase training framework for segmentation and domain adaptation

Loss Component	**Type**	**Applied To**	**Weight (λ)**
Dice Loss (weighted)	Per-region	HGG + LGG (seg)	1
Focal Loss	BCE	HGG + LGG (seg)	0.5
Radiomics Loss (source)	MSE	HGG only	0.1 (ep$\leq$5), 0.2 (ep>5)
Radiomics Consistency (target)	MSE	LGG only	0.0 (ep$\leq$10), 0.0001 (ep>10)
Adversarial Loss	BCE	LGG only	0.001
Discriminator Loss	BCE	HGG + LGG	0.5$\times$ ($src + tgt$)

enables the network to distinguish between source and target domain features, ensuring stable adversarial training throughout adaptation.

4.5 Memory Footprint

As detailed in Table 10, the primary backbone, UNet3D, contains 1.4 million parameters, accounting for over 98% of the total model size, and all parameters are trainable. The supporting modules, RadRegressor and DomainDiscriminator, are extremely compact, with only 12.9K and 10.4K parameters, respectively. Overall, the complete system comprises approximately 1.43 million trainable parameters, with no frozen layers, ensuring full adaptability during training while maintaining a compact computational footprint.

As shown in Table 11, despite processing volumetric 3D data, the model requires only 115.9 GFLOPs, which is low for a 3D segmentation network. Each parameter contributes roughly 82.7K operations. Additionally, the memory requirements are minimal, with 5.61 MB in FP32 precision and 2.81 MB in FP16. This indicates that the model can be deployed on standard hardware without excessive memory overhead.

Table 10: Detailed parameter distribution across model components, including total, trainable, and non-trainable parameters of the complete system.

Component	Total Parameters	Trainable Parameters	Non-train. Parameters
UNet3D	1,401,891 (1.40 M)	1,401,891 (1.40 M)	0
RadRegressor	12,936 (12.9 K)	12,936 (12.9 K)	0
DomainDiscrim.	10,369 (10.4 K)	10,369 (10.4 K)	0
Total	1,425,196 (1.43 M)	1,425,196 (1.43 M)	0

Table 11: Computational requirements of the model, including total FLOPs, operations per parameter, and memory usage in FP32 and FP16 precision.

Metric	Value
FLOPs	115,895,959,552 (∼115.90 GFLOPs)
FLOPs per Parameter	82,671.16 (∼82.67K)
Memory (FP32)	5.61 MB
Memory (FP16)	2.81 MB

We have further measured wall-clock inference time and peak GPU memory. Table 12 reports the results across different batch sizes, along with a CPU baseline at a batch size of 1. At batch size 1, GPU inference takes 171.58 ms on average, with a standard deviation of 2.61 ms. CPU inference takes 3386.4 ms on average at the same batch size. The GPU therefore gives roughly a 20-fold speedup over CPU.

Throughput begins at 5.83 volumes per second at batch size 1 and drops slightly to 5.40 volumes per second at batch size 4. Variance in latency also grows with batch size. The standard deviation rises from 2.61 ms at batch size 1 to 11.96 ms at batch size 4. Additionally, peak GPU memory scales linearly with batch size. A full training step, which adds the backward pass and the optimizer update, reaches 2.49 GB, 4.96 GB, and 9.87 GB at the same batch sizes.

Table 12: Measured inference latency, throughput, and peak GPU memory across different batch sizes (denoted as BS). CPU latency is reported only at batch size 1 as a secondary deployment data point.

Metric	BS-1	BS-2	BS-4	CPU
Mean Latency (ms)	171.58	357.59	740.83	3386.4
Std (ms)	2.61	5.4	11.96	30.33
Min (ms)	166.99	348.75	713.70	3345.33
Max (ms)	176.72	364.97	754.13	3451.55
Throughput (vol/s)	5.828	5.593	5.399	0.295
Inference Peak (MB)	1144.43	2236.98	4384.59	–
Fwd+Bwd Peak (MB)	2490.75	4965.26	9878.89	–
Full Step Peak (MB)	2490.75	4959.45	9873.08	–

4.6 Ablation Study

To quantify the contribution of each proposed component, we conduct a structured ablation study on BraTS 2020. The study is organized into three groups, each isolating one core component: the Radiomics Consistency Bridge, the Self-Feedback mechanism, and the Domain Alignment module. Within each group, configurations are built up progressively from a component-absent baseline. Table 13 summarizes the results.

The Radiomics Bridge group shows that supervised consistency ($\mathcal{L}_r^{\text{sup}}$) provides the larger individual gain of +5.79%, while unsupervised self-consistency ($\mathcal{L}_r^{\text{self}}$) contributes +3.46%. Both modes together yield +8.67%, confirming complementary supervision signals. In the Self-Feedback group, the absence of a warm-up schedule (S1) produces the highest variance of 0.81, while progressive addition of IoU gating and the $\beta^{\max}$ ceiling each reduce variance and improve performance, with S4 achieving a gain of +7.51%. In the Domain Alignment group, CORAL alone contributes +4.20% and DANN alone contributes +7.79%, with their combination in the full model reaching 62.36%. Across all groups, the standard deviation decreases monotonically from 0.81 to 0.43.

Beyond component contributions, we examine the sensitivity of RADIANT to the loss weights λ_{rad} and λ_{adv}, and the GRL scaling schedule α, as illustrated in Figure 6. For the loss weights, performance degrades at both extremes, where a smaller λ_{rad} underconstrains the bottleneck and a larger one disrupts segmentation gradients. A similar pattern holds for λ_{adv}. For the GRL schedule, fixed α configurations underperform linear alternatives, and overly aggressive schedules ($1.0 \to 3.0$) also reduce performance. The chosen configuration (Linear $1.0 \to 2.0$) achieves the highest Avg DSC of 62.36%, validating the selected hyperparameters.

4.7 Performance comparison

Table 14 compares our model against existing LGG domain adaptation methods across BraTS datasets. On BraTS18,

Table 13: Component-wise ablation configurations. Std is the standard deviation across five independent runs. ΔDSC% is the gain relative to each group's own baseline row. $\uparrow$ denotes higher is better.

Group	**Exp.**	**Configuration**	**WT**	**TC**	**ET**	**Avg DSC** $\uparrow$	**Std** $\downarrow$	**ΔDSC%** $\uparrow$
Radiomics Bridge	R1	No RadRegressor	68.31	39.14	34.72	47.39	0.71	0.00
	R2	RadRegressor, $\mathcal{L}_r^{\text{sup}}$ only	73.45	44.87	41.23	53.18	0.65	+5.79
	R3	RadRegressor, $\mathcal{L}_r^{\text{self}}$ only	71.12	42.53	38.91	50.85	0.68	+3.46
	R4	Both modes, no self-feedback or alignment	76.83	47.62	43.74	56.06	0.57	+8.67
Self-Feedback	S1	No warm-up, fixed β_{self} from epoch 1	73.27	44.51	40.83	52.87	0.81	0.00
	S2	Warm-up only, no IoU gating	76.14	47.93	44.62	56.23	0.69	+3.36
	S3	Warm-up + IoU gating, no $\beta^{\max}$ cap	78.49	50.87	47.11	58.82	0.61	+5.95
	S4	Warm-up + IoU gating + $\beta^{\max}$ cap	80.13	52.34	48.67	60.38	0.52	+7.51
Domain Alignment	D1	No alignment	70.84	41.27	37.63	49.91	0.74	0.00
	D2	CORAL only, no DANN	74.53	45.61	42.18	54.11	0.66	+4.20
	D3	DANN only, no CORAL	77.92	49.34	45.83	57.70	0.59	+7.79
RADIANT	Full	All components combined	81.98	54.21	50.89	62.36	0.43	—

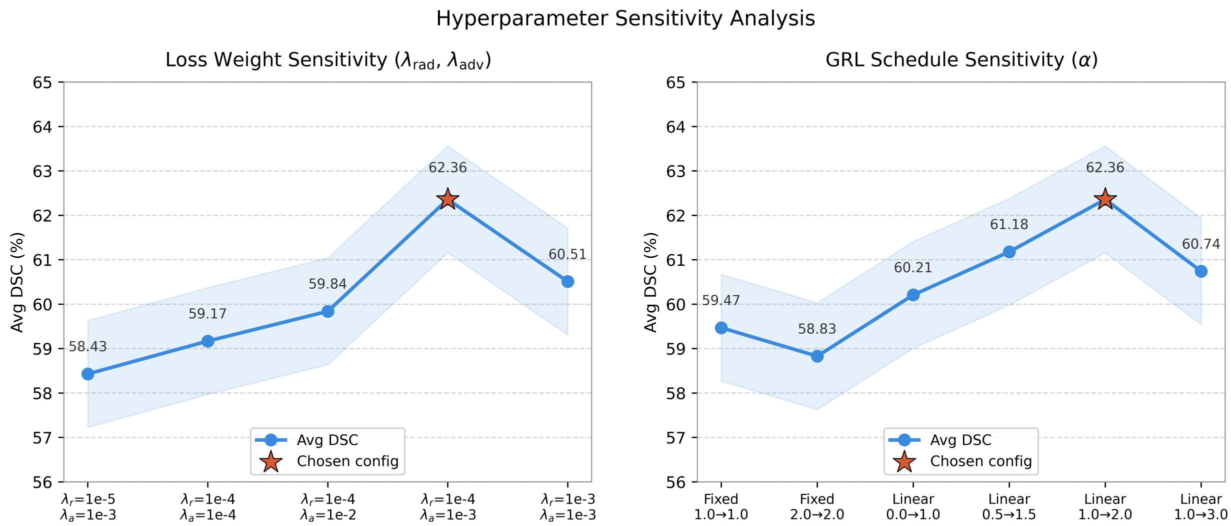

Figure 6: Sensitivity analysis of loss weights (λ_{rad}, λ_{adv}) and GRL scaling schedule (α) on BraTS 2020. Stars denote the chosen configuration.

RADIANT achieves lower WT DSC than UDA-SE [15] but substantially higher TC and ET scores. Against BBUDA [14], RADIANT outperforms across all three subregions. On BraTS19, DS-Adv [8] achieves higher WT and ET scores, while RADIANT records the highest TC DSC among all compared methods. These results suggest that RADIANT trades marginal WT performance for considerably stronger TC segmentation, which captures the active and necrotic tumor core. On BraTS20, RADIANT maintains consistent TC and ET performance across all three datasets.

To compare our findings under a more controlled setting, we reproduced DS-Adv [8] and UDA-SE [15] under similar experimental conditions as RADIANT. Table 15 shows the comparison on a 70:30 data split, with a $128 \times 128 \times 128$ patch size and z-score normalization on the BraTS 2020 dataset. Both baselines were re-implemented using their original backbone architectures and training objectives and were further evaluated by introducing correlation alignment (CORAL) and radiomics consistency into each framework. Both baselines improve as each component is added, yet

Table 14: Comparison of subregion DSC scores (WT, TC, ET) on BraTS datasets for LGG domain adaptation methods reported in the existing literature. The reported results are obtained directly from the source manuscripts.

Models	Dataset	Subregion DSC (%)		
		WT	TC	ET
DS-Adv [8]	BraTS19	88.85	48.31	63.63
UDA-SE [15]	BraTS18	84.11	47.11	32.67
BBUDA [14]	BraTS18	82.14	31.02	46.13
RADIANT (Ours)	BraTS18	83.12	55.34	51.87
	BraTS19	84.53	57.62	54.94
	BraTS20	81.98	54.21	50.89

their strongest configurations achieve Avg DSC of 57.84% and 55.61%, respectively, remaining below RADIANT's 62.36%. The higher standard deviations further indicate less stable training compared to RADIANT.

Table 15: Controlled baseline comparison on BraTS 2020 under similar experimental conditions. DSC values are averaged over five independent runs. **CORAL**, and **Rad.** denotes domain **cor**relation **al**ignment, and **rad**iomics respectively.

Method	CORAL	Rad.	DSC (%) ↑	ΔStd ↓
DS-Adv [8]	✗	✗	49.17	2.73
	✓	✗	53.42	2.61
	✗	✓	51.83	2.57
	✓	✓	57.84	2.51
UDA-SE [15]	✗	✗	47.63	1.89
	✓	✗	51.27	1.81
	✗	✓	49.94	1.76
	✓	✓	55.61	1.64
RADIANT	✓	✓	62.36	0.43

5 Discussion

We propose RADIANT, a structure-preserving domain adaptation framework for 3D brain tumor segmentation that enables a model trained on HGG data to generalize effectively to LGG without requiring LGG annotations. The framework is built upon a lightweight 3D U-Net-based hierarchical feature learning architecture enhanced with instance normalization to enhance robustness against domain shifts. Extended skip connections are integrated to maintain precise tumor boundary representation, while the bottleneck serves as a shared latent space supporting multiple learning objectives. A radiomics-guided consistency strategy is introduced to connect deep feature representations with global tumor characteristics and maintain anatomical consistency. This strategy combines radiomics feature extraction with a lightweight regression module linked to bottleneck features, encouraging structural consistency across domains.

In addition, a dual-mode consistency mechanism is incorporated to reduce the influence of unreliable predictions during early training and gradually impose stronger consistency constraints as learning stabilizes. The framework further incorporates adversarial domain alignment at the bottleneck to reduce discrepancies between source and target feature distributions, which helps prevent disruption of low-level features that capture modality-specific intensity patterns while promoting domain invariance.

We evaluate the proposed model on three benchmark datasets: BraTS 2018, BraTS 2019, and BraTS 2020. On BraTS 2018, it achieves DSC scores ranging from 80.17% to 86.91% for WT, 54.73% to 57.75% for TC, and 48.01% to 59.24% for ET. On BraTS 2019, the corresponding DSC ranges are 82.77% to 89.77% (WT), 53.97% to 64.51% (TC), and 48.09% to 58.19% (ET). For BraTS 2020, the model attains DSC scores ranging from 76.21% to 85.52% (WT), 52.91% to 58.36% (TC), and 47.08% to 53.80% (ET). Across all datasets, the highest segmentation performance is observed when the source and target modalities match, highlighting the effectiveness of within-modality domain adaptation across all tumor regions.

The training follows a two-phase approach, beginning with supervised pretraining on HGG data and then proceeding with unsupervised adaptation to LGG, using separate learning rates for the encoder and decoder. To promote domain-invariant representations, the framework incorporates adversarial training with a discriminator and a gradient reversal factor that increases linearly over time. Optimization combines Dice and Focal losses for segmentation with a radiomics loss on the source domain to align features with ground-truth radiomics and radiomics consistency on the target domain to maintain self-consistency using pseudo-radiomics features. L2 regularization is applied throughout both phases to prevent overfitting and guide alignment of deep feature representations across tumor domains.

Compared to recent SOTA methods reported in the existing literature, RADIANT consistently achieves balanced and high DSC across all subregions in the three benchmark datasets. On BraTS18, it surpasses BBUDA [14] and UDA-SE [15], with particularly strong TC and ET performance. On BraTS19, RADIANT attains the highest TC score while maintaining competitive WT and ET results, outperforming DS-Adv [8] in subregion balance. On BraTS20, it continues to achieve stable TC and ET accuracy, demonstrating its applicability across multiple datasets.

Despite its strong performance, RADIANT has several limitations. First, it relies on high-quality annotated HGG data for pretraining, which may limit its applicability in scenarios where such labels are scarce. Second, the BraTS benchmark contains considerably fewer LGG cases than HGG cases, which constrains the diversity of adaptation signals available during unsupervised training and may limit the robustness of the learned target representations. Third, the model is developed and evaluated solely on MRI scans, leaving its performance on computed tomography (CT) and positron emission tomography (PET) untested. Finally, while extensive evaluations are conducted on the BraTS datasets, the model's generalizability to more diverse clinical data from different scanners or protocols remains to be established.

6 Conclusion

We introduce RADIANT, a robust framework for brain MRI domain adaptation that achieves consistent DSC performance across the BraTS 2018, BraTS 2019, and BraTS 2020 benchmark datasets. Across different adaptation settings, RADIANT achieves DSC scores ranging from 80.17% to 86.91% for WT, 54.73% to 57.75% for TC, and 48.01% to 59.24% for ET on BraTS 2018. On BraTS 2019, the corresponding DSC ranges are 82.77% to 89.77% (WT), 53.97% to 64.51% (TC), and 48.09% to 58.19% (ET). For BraTS 2020, the model attains DSC scores ranging from 76.21% to 85.52% (WT), 52.91% to 58.36% (TC), and 47.08% to 53.80% (ET). These results demonstrate that RADIANT provides balanced and robust subregion performance across multiple datasets and effectively adapts knowledge from HGG to LGG without requiring target-domain annotations.

The model employs a lightweight 3D U-Net architecture with a dual-mode consistency mechanism to preserve tumor boundaries accurately and mitigate noisy predictions. A key strength of RADIANT is its radiomics-guided consistency strategy, which combines radiomics feature extraction with a lightweight regression module to connect deep feature representations to global tumor characteristics to ensure anatomical consistency. Furthermore, the framework incorporates adversarial domain alignment to achieve domain invariance by reducing discrepancies between feature distributions across domains.

Despite its strong performance, RADIANT may be limited when annotated HGG data are scarce. Its generalizability to other modalities and diverse clinical datasets remains uncertain. The proposed RADIANT advances domain adaptation from HGG to LGG, supporting radiologists in making accurate decisions and improving patient outcomes without requiring target-domain labels.

Statements and Declarations

Funding: This research was funded by the Institute for Advanced Research (IAR), United International University, under Project Code: UIU-IAR-02-2023-SE-23.

Ethical Approval: Not applicable. This study used publicly available BraTS datasets and did not involve direct interaction with human participants or access to patient-identifying information.

Competing interests: The authors state that they have no known financial conflicts of interest or personal relationships that could have influenced the work presented in this paper.

Data Availability: In this study, we use three publicly available datasets: the BraTS benchmark datasets from 2018 to 2020 [21, 22, 23].

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